

Data Management Layer Changes — Executive Summary

This report documents changes applied to the watch-status persistence layer in Mediaflix. Changes include enabling SQLite WAL mode, adding an index on the `watched` column, and introducing an in-memory LRU cache to reduce reads to disk.

The following pages summarize the changes, benchmark methodology, results, and recommendations.

What Was Changed

- Enabled SQLite WAL journal mode for better concurrency and faster reads.
- Created index: idx_watch_status_watched ON watch_status(watched).
- Changed column types to INTEGER/REAL for compact storage and deterministic queries.
- Implemented a thread-safe in-memory LRU cache (default 2000 entries) with write-through updates.
- Added chunked batch queries to avoid SQLite parameter limits when checking many files.

Benchmark: Method and Results

Warm-fill time (cache-enabled): 0.5458 s

Lookup time (cache-enabled, 2000 lookups): 1.0600 s

Warm-fill time (cold manager): 0.6412 s

Lookup time (cold manager, 2000 lookups): 1.2315 s

Notes:

- "Warm-fill" measures the initial pass to populate cache or touch DB rows.
- The cache-enabled manager typically shows lower elapsed lookup times because repeated lookups hit memory.

Recommendations and Next Steps

- Tune `\_cache\_max` based on available memory and workload. For many desktop installs, 2000 keys is conservative.
- Consider periodic background DB VACUUM and ANALYZE to keep indexes optimized for large/long-lived DBs.
- For multi-process use, move to a small local server DB (Postgres/SQLite with a server) or shared cache (Redis).
- Add simple instrumentation (timings counters) in production to monitor cache hit rates and query times.